



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

July 11, 2025

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Introduce the concept of cross-validation for model evaluation and describe at least two common strategies for implementing it, including their advantages and disadvantages.

Exercise 2 (7 marks)

Illustrate and explain the Bayesian interpretation of ridge regression.

Exercise 3 (2 marks)

Consider the following snippet of code:

```
1 def ker_1(x_1, x_2):  
2     return -np.dot(x_1-x_2, x_1-x_2)  
3  
4 def ker_2(x_1, x_2):  
5     return np.exp(ker_1(x_1, x_2))  
6  
7 SVM_model_1 = svm.SVC(kernel=ker_1)  
8 SVM_model_1.fit(X_train, t_train)  
9 prediction_1 = SVM_model_1.predict(X_test)  
10  
11 SVM_model_2 = svm.SVC(kernel=ker_2)  
12 SVM_model_2.fit(X_train, t_train)  
13 prediction_2 = SVM_model_2.predict(X_test)  
14  
15 print(accuracy_score(t_test, prediction_2) - accuracy_score(t_test, prediction_1))
```

1. What is the aim of this code? What methods are being used?
2. Are there any problems with the solution implemented here? If so, propose a fix.

1. The code is trying to solve a classification problem using support vector machines with two different kernels, comparing their accuracy
2. The first kernel is not a valid kernel, in fact it is always negative (or zero)! However, the second kernel is a Gaussian kernel. Even removing the minus in front of the kernel function won't make the kernel valid. A possible fix, would be to use a simpler kernel, like a linear one.

Exercise 4 (2 marks)

Tell if the following statements are true or false. Motivate your answers.

1. In consistent learning, the training error is an accurate estimator of the true error of the model.
2. The test error is an accurate estimator of the true error of the model only if the VC dimension is small.
3. The larger the VC dimension, the more the hypothesis space is prone to overfitting.
4. The more the samples in the training set, the more the training error is an accurate estimator for the true error of the model.

1. **False.** The training error is always biased even more so in consistent learning in which it is always zero.
2. **False.** The test error is always unbiased and its reliability depends on the size of the test set only, not on the VC dimension.
3. **True.** The VC dimension is an index of the complexity of the hypothesis space and, thus, quantifies the variance.
4. **True.** In all cases (consistent and agnostic learning) the accuracy of the training error increases as the number of training samples increases.

Exercise 5 (2 marks)

Tell if the following statements about multi-armed bandits are true or false. Motivate your answers.

1. UCB1 is a deterministic algorithm
2. The expected total regret of UCB1 after T rounds is upper bounded by a constant independent of T .
3. Exploration is trivial in multi-armed bandit problems since there is only one state.
4. Thompson sampling needs to sample multiple values at each round.

1. **True.** Given the same sequence of rewards, UCB1 will always play the same sequence of actions.
2. **False.** The regret grows logarithmically with T .
3. **False.** Multi-armed bandit problems are a prime example of the exploration-exploitation dilemma: gather more information from less explored options or select the option that seems the best according to current knowledge.
4. **True.** Thompson sampling samples N values at each round, one for each arm, from the arms' posteriors.

Exercise 6 (2 marks)

You are a machine learning expert volunteering for the organizing committee of the next COLT (Conference on Learning Theory). Consider the following two challenges:

1. **Talk Attendance Prediction:** To help with room assignment, the committee wants to estimate how many people will attend each talk. You are given historical data from past COLT conferences: each talk is described by topic area, author affiliations, keywords, and time slot. The actual attendance numbers are also recorded.
2. **Poster Session Grouping:** The committee wants to organize accepted posters into coherent sessions. Each poster is represented by its title, abstract, author list, and features describing the content of the poster (e.g., text embedding from a language model). Your task is to group papers into sessions such that papers within the same session are thematically similar.

Model the two problems described above as ML problems, specifying the data (features and target or states, actions, and reward), the problem class. **For the first problem only**, what approach would you use if the committee wants to have a measure of the uncertainty of the prediction too?

1. SUPERVISED LEARNING, REGRESSION since we are to predict a numerical variable. Input: topic area, authors affiliations, keywords, time slot. Output: number of attendees. To have a measure of the uncertainty of the prediction, we can use, for instance, GPs or BLR.
2. UNSUPERVISED LEARNING, CLUSTERING since we need to group points that share similarities. Input: title, abstract, author list, paper embeddings.

Exercise 7 (4 marks)

Consider an MDP with 3 states A, B, C , 2 actions x, y , and discount factor $\gamma = 0.5$. Running policy π , we have collected the following trajectories, displayed as sequences of (state, action, reward) tuples:

$$(A, x, 1) \rightarrow (B, y, 2) \rightarrow (B, x, 4) \rightarrow C$$

$$(A, x, 1) \rightarrow (B, x, 0) \rightarrow C$$

Answer the following questions showing all the computations. Make and state any assumption you deem necessary to answer.

1. Estimate V^π using first-visit Monte Carlo
2. Estimate V^π using every-visit Monte Carlo
3. Estimate V^π for all states using TD(0) with learning rate $\alpha = 0.5$
4. Provide a policy that is compatible with the observed trajectories

We will consider C as a terminal state, so $V^\pi(C) = 0$ for all the methods. For TD, we will assume the initial estimates are all zero.

$$1. V^\pi(A) = \frac{(1 + \frac{1}{2} * 2 + \frac{1}{4} * 4) + (1 + \frac{1}{2} * 0)}{2} = 2$$

$$V^\pi(B) = \frac{(2 + \frac{1}{2} * 4) + 0}{2} = 2$$

2. $V^\pi(A)$: same as first-visit

$$V^\pi(B) = \frac{(2 + \frac{1}{2} * 4) + 4 + 0}{3} = \frac{8}{3}$$

$$3. V(A) = 0 + \frac{1}{2}(1 + \frac{1}{2} * 0 - 0) = \frac{1}{2}$$

$$V(B) = 0 + \frac{1}{2}(2 + \frac{1}{2} * 0 - 0) = 1$$

$$V(B) = 1 + \frac{1}{2}(4 + \frac{1}{2} * 0 - 1) = \frac{5}{2}$$

$$V(A) = \frac{1}{2} + \frac{1}{2}(1 + \frac{1}{2} * \frac{5}{2} - \frac{1}{2}) = \frac{11}{8}$$

$$V(B) = \frac{5}{2} + \frac{1}{2}(0 + \frac{1}{2} * 0 - \frac{5}{2}) = \frac{5}{4}$$

4. A possible policy is given by $\pi(x|A) = 1$ and $\pi(x|B) = \pi(y|B) = \frac{1}{2}$.

Exercise 8 (4 marks)

You are training a binary classifier using **ℓ_2 -regularized logistic regression**. The model predicts the probability that a binary label $y \in \{0, 1\}$ is 1 given input $x \in \mathbb{R}^d$, using the sigmoid function: $\sigma(z) = \frac{1}{1+e^{-z}}$. The regularized loss function for a single sample (x, y) is (the bias term is zero):

$$\ell(w; x, y) = -y \log \sigma(w^\top x) - (1 - y) \log(1 - \sigma(w^\top x)) + \frac{\lambda}{2} \|w\|^2$$

where $\lambda > 0$ is the regularization parameter.

1. Derive the gradient $\nabla_w \ell(w; x, y)$ of the loss function with respect to the weight vector w .
2. Assume the current weight vector is $w = [1, -1]^\top$, and you receive the sample $x = [2, 3]^\top$ and $y = 1$. Use a learning rate $\eta = 1$ and regularization parameter $\lambda = 1/2$. Compute the stochastic gradient and update the weights.
3. Find a point $x \in \mathbb{R}^2$ that is classified differently by the updated model compared to the original model.

Hint: $\sigma(-1) \approx 1/4$.

1. Let $z = w^\top x$, and $\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$. The loss can be rewritten as:

$$\ell(w) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y}) + \frac{\lambda}{2} \|w\|^2.$$

With standard computations, the gradient is:

$$\nabla_w \ell(w; x, y) = (\sigma(w^\top x) - y)x + \lambda w.$$

2. We apply stochastic gradient descent on the sample:

$$w' = w - \eta \nabla_w \ell(w; x, y) = [1, -1]^\top - (\sigma(-1) - 1)[2, 3]^\top - \frac{1}{2}[1, -1]^\top \quad (1)$$

$$\approx [1, -1]^\top - [-3/2, -9/4]^\top - [1/2, -1/2]^\top = [2, 7/4]^\top. \quad (2)$$

3. We have to find a point $x = [x_1, x_2]^\top$ such that $(w^\top x)((w')^\top x) < 0$, which leads to the inequality:

$$(w^\top x)((w')^\top x) = (x_1 - x_2)(2x_1 + 7/4x_2) < 0. \quad (3)$$

For instance, we pick $x = [0, 1]^\top$.

